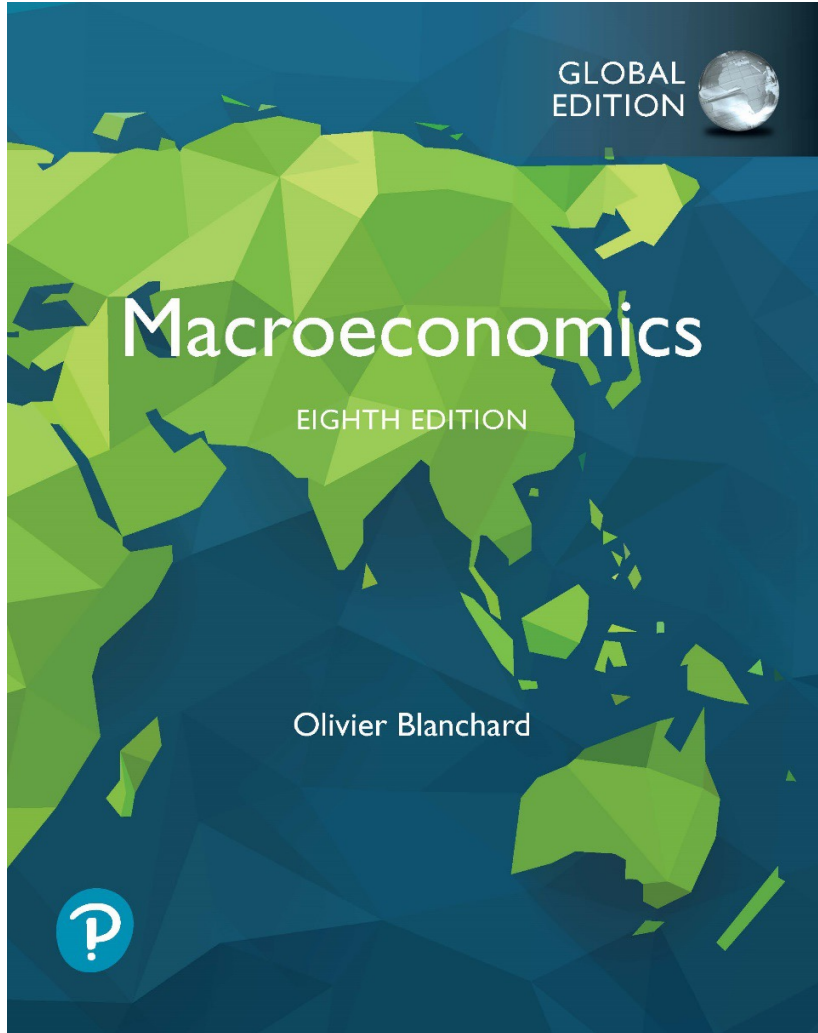


Macroeconomics

Eighth Edition, Global Edition



Chapter 5

Goods and Financial Markets: The
IS-LM Model

Chapter 5 Outline

Goods and Financial Markets: The *IS-LM* Model

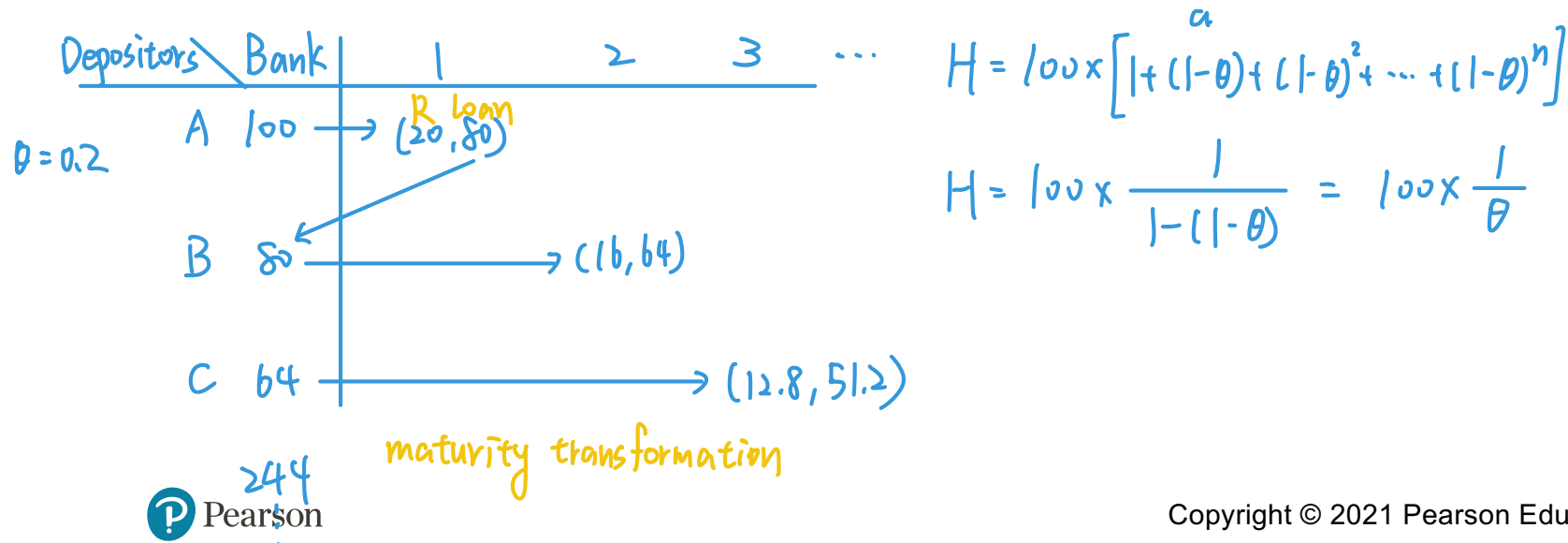
5.1 The Goods Market and the *IS* Relation

5.2 Financial Markets and the *LM* Relation

5.3 Putting the *IS* and the *LM* Relations Together

5.4 Using a Policy Mix

5.5 How Does the *IS-LM* Model Fit the Facts?



Goods and Financial Markets: The *IS-LM* Model

- We looked at the goods market in Chapter 3, and financial markets in Chapter 4.
- In this chapter, we look at goods and financial markets together, and understand how output and the interest rate are determined in the short run.
- John Hicks and Alvin Hansen called this framework the *IS-LM* model.

$$H = C + R = \alpha D + \theta D = (\alpha + \theta)D \quad \text{--- ①}$$

$$M = C + D = \alpha D + D = (\alpha + 1)D \quad \text{--- ②}$$

$$\text{Assume } C = \alpha D \quad R = \theta D \quad \alpha < 1, \theta < 1$$

$$\text{②/①} \Rightarrow M = \underbrace{\left(\frac{\alpha+1}{\alpha+\theta} \right)}_{>1} H \quad m = \frac{\alpha+1}{\alpha+\theta} > 1$$

$$\frac{\partial m}{\partial \theta} < 0, \quad \frac{\partial m}{\partial \alpha} < 0$$

$$\alpha \text{ 愈大 } D \text{ 愈小 } \text{ b/c } M = C + D \quad (C = \alpha D)$$

$$\text{if } \alpha = 0 \Rightarrow M = \frac{1}{\theta} H \Rightarrow \theta M = H$$

$$\theta = 1 \Rightarrow M = H$$

5.1 The Goods Market and the I/S Relation (1 of 6)

- In the model developed in Chapter 3, investment was assumed to be constant for simplicity.
- In fact, investment depends on production Y (or sales) and the interest rate i .

$$I = I(Y, i) \quad (5.1)$$

(+, -)

- So equilibrium in the goods market becomes:

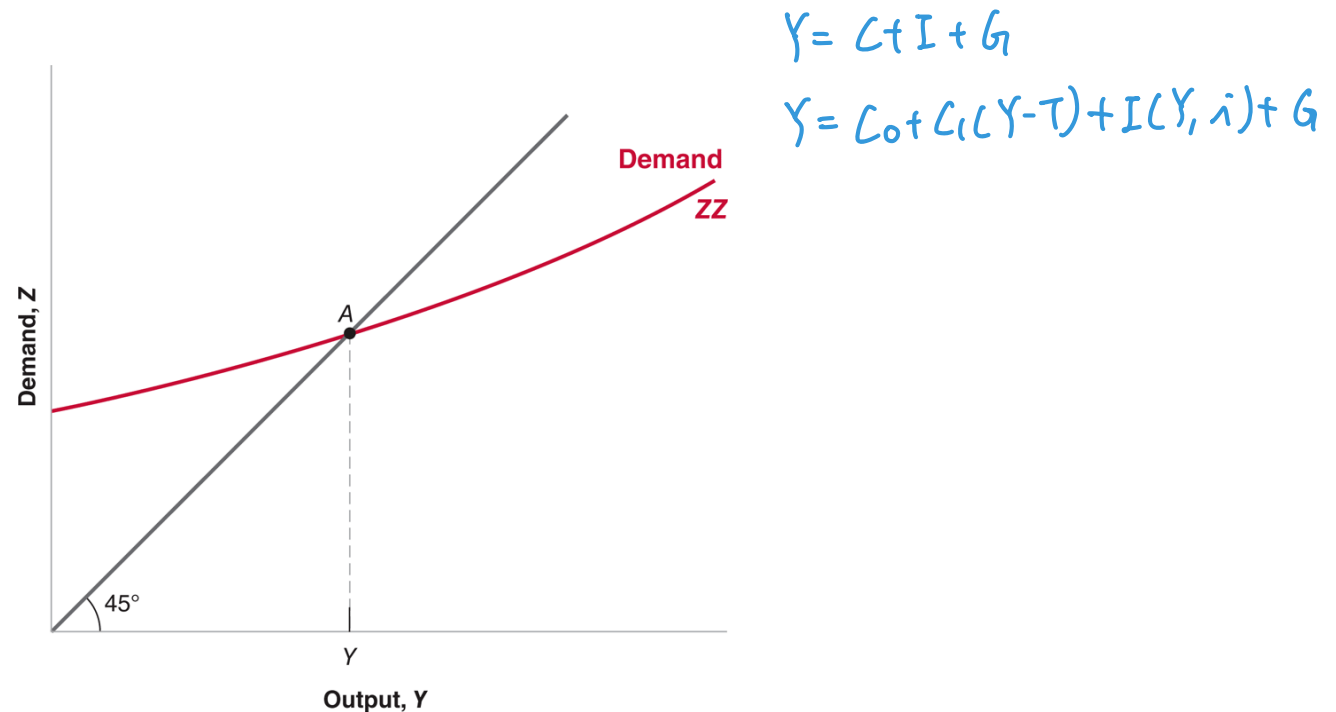
$$Y = C(Y - T) + I(Y, i) + G \quad (5.2)$$

which is the I/S relation.

5.1 The Goods Market and the /S Relation (2 of 6)

Figure 5.1 Equilibrium in the Goods Market

The demand for goods is an increasing function of output. Equilibrium requires that the demand for goods be equal to output.



5.1 The Goods Market and the *IS* Relation (3 of 6)

- *ZZ* is upward-sloping because, for a given value of the interest rate, an increase in output leads to an increase in the demand for goods through its effects on consumption and investment.
- *ZZ* is flatter than the 45-degree line because we have assumed that an increase in output leads to a less than one-for-one increase in demand.
- The intersection of *ZZ* and the 45-degree line (point *A*) is the equilibrium level of output.

5.1 The Goods Market and the IS Relation (4 of 6)

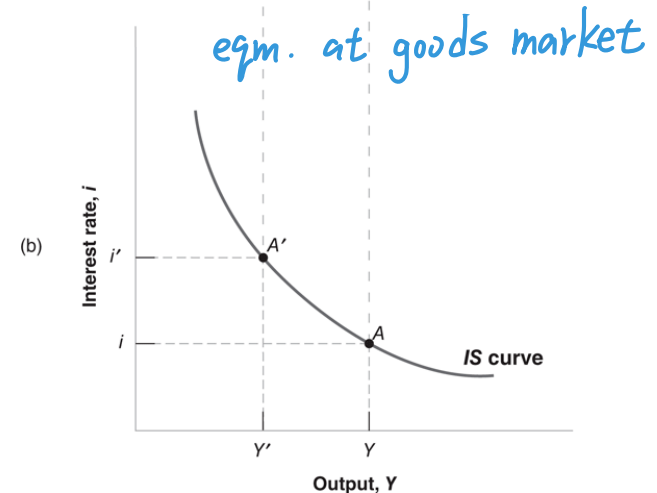
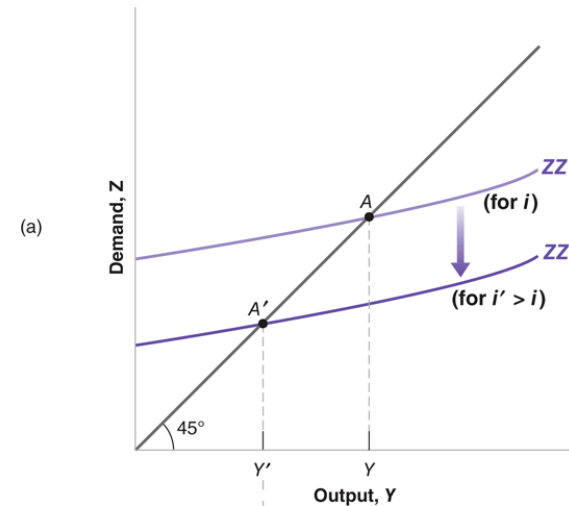
Figure 5.2 The *IS* Curve

(a) An increase in the interest rate decreases the demand for goods at any level of output, leading to a decrease in the equilibrium level of output.

(b) Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.

The **IS curve** is therefore downward sloping.

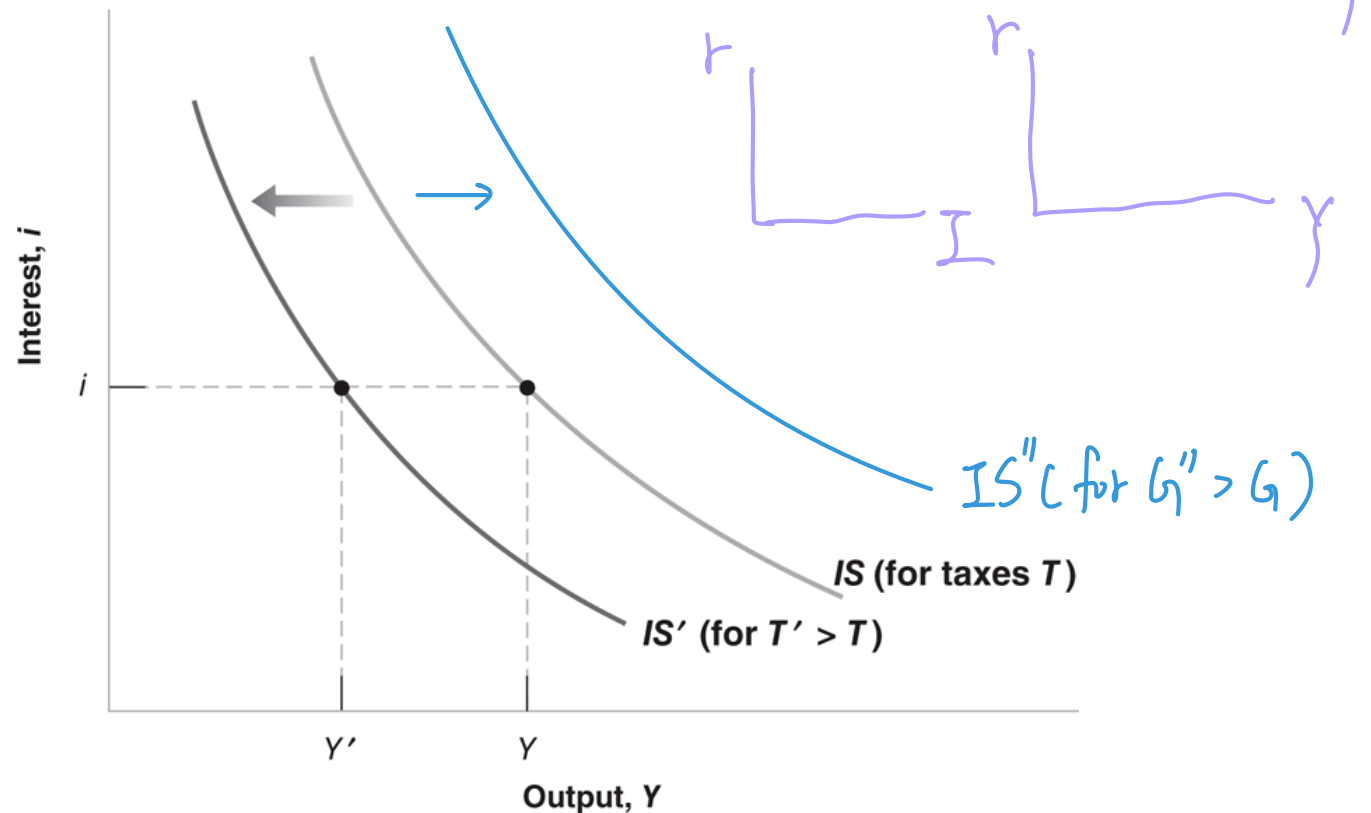
Goods Market : $IS \rightarrow Y = C(Y-T) + I(Y, i) + G$
Given i, G, T



5.1 The Goods Market and the *IS* Relation (5 of 6)

Figure 5.3 Shifts of the *IS* Curve

An increase in taxes shifts the *IS* curve to the left.



5.1 The Goods Market and the IS Relation (6 of 6)

- Downward-sloping IS curve: Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.
- Shifting the IS curve: Changes in factors that decrease (increase) the demand for goods given the interest rate shift the IS curve to the left (right).

5.2 Financial Markets and the *LM* Relation (1 of 3)

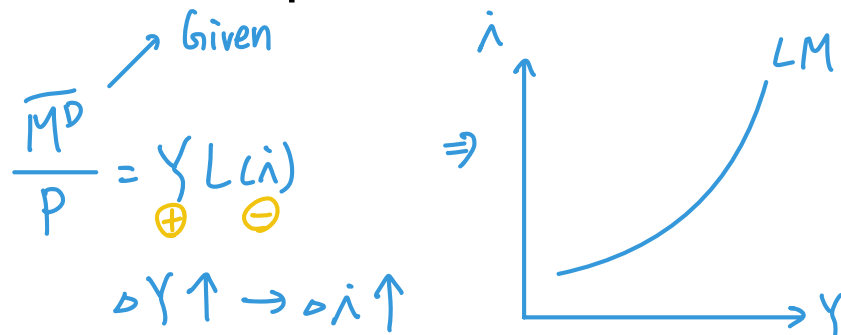
- Recall Chapter 4: $M = PY L(i)$
- Divide both sides of the equation by the price level P :

$$\frac{M}{P} = YL(i) \quad (5.3)$$

real money demand

which is the *LM* relation.

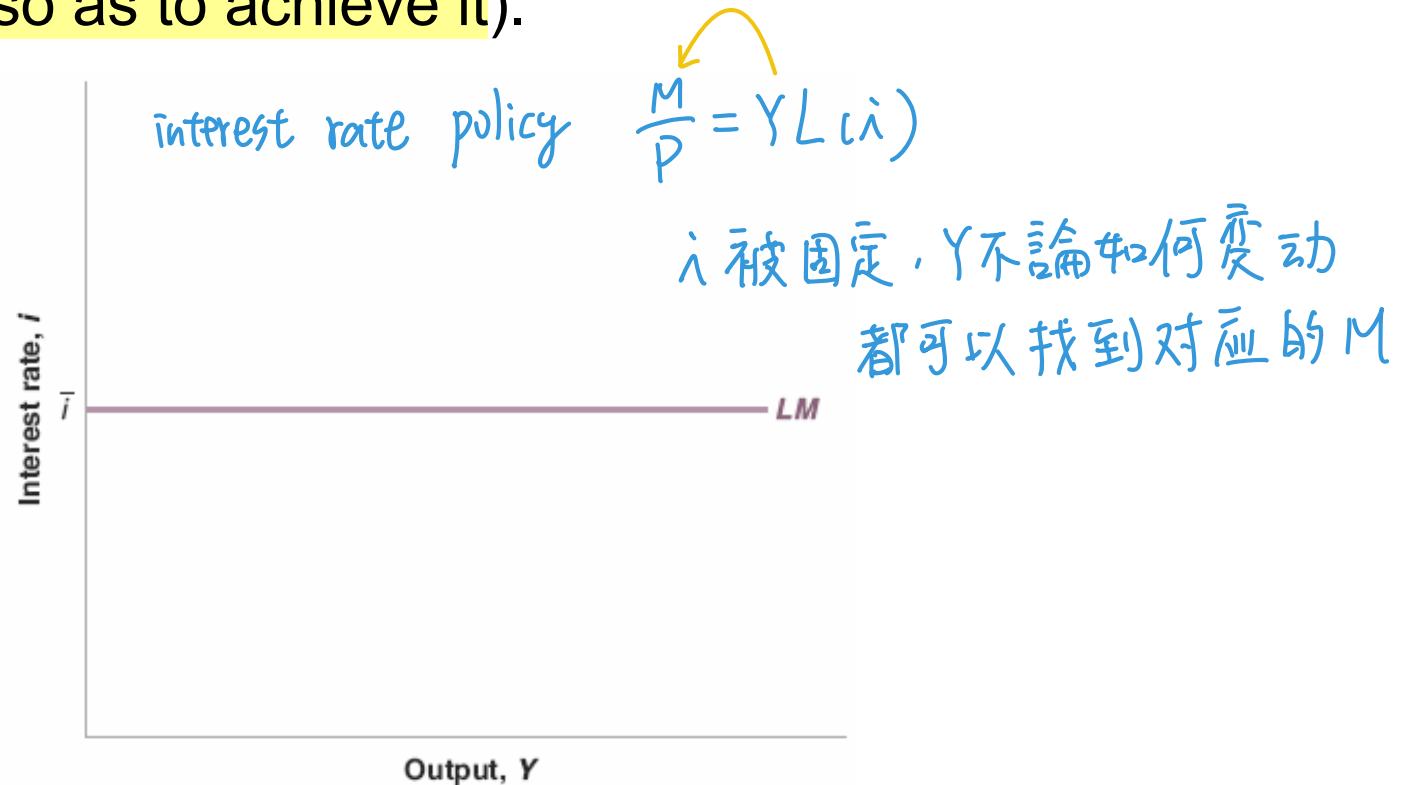
- In equilibrium, real money supply equals the real money demand, which depends on real income Y , and the interest rate i .



5.2 Financial Markets and the *LM* Relation (2 of 3)

Figure 5.4 The *LM* Curve

The central bank chooses the interest rate (and adjusts the money supply so as to achieve it).



5.2 Financial Markets and the *LM* Relation (3 of 3)

- *IS* relation: $Y = C(Y-T) + I(Y, i) + G$
- *LM* relation: $i = \bar{i}$
- The *IS* and *LM* relations together determine output.
- *Any point* on the downward sloping *IS* curve corresponds to equilibrium in the goods market.
- *Any point* on the horizontal *LM* curve corresponds to equilibrium in financial markets.
- *Only at* their intersection (*point A*) are both equilibrium relations satisfied.

5.3 Putting the *IS* and the *LM* Relations Together (1 of 6)

$$m = \frac{M}{P} \quad (+A)$$

$$L = L_1(Y) + L_2(r) = kY - hr$$

$$IS: Y = C(Y - T) + I(Y, \bar{r}) + G$$

$$LM: \frac{M}{P} = Y L(\bar{r})$$

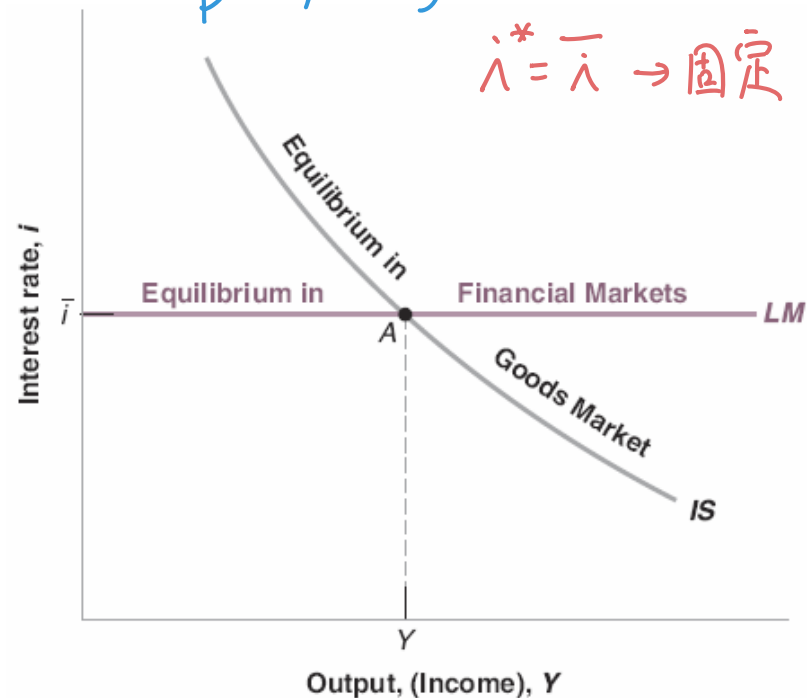
$$\bar{r}^* = \bar{r} \rightarrow \text{固定}$$

Figure 5-5 The *IS–LM* Model

Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.

This is represented by the *IS* curve.

Equilibrium in financial markets is represented by the horizontal *LM* curve. Only at point *A*, which is on both curves, are both goods and financial markets in equilibrium.



5.3 Putting the *IS* and the *LM* Relations Together (2 of 6)

- Fiscal Policy:

Decrease in $G-T \Leftrightarrow$ **fiscal contraction** $\Leftrightarrow$ **fiscal consolidation**

Increase in $G-T \Leftrightarrow$ **fiscal expansion**

Budget balance : $G-T$

$$Y = AE = C + I + G = \alpha + \beta(Y - T) + (e - dr) + G$$
$$Y = \frac{\alpha + e + G - \beta T}{1 - \beta} - \frac{d}{1 - \beta} d$$

5.3 Putting the *IS* and the *LM* Relations Together (3 of 6)

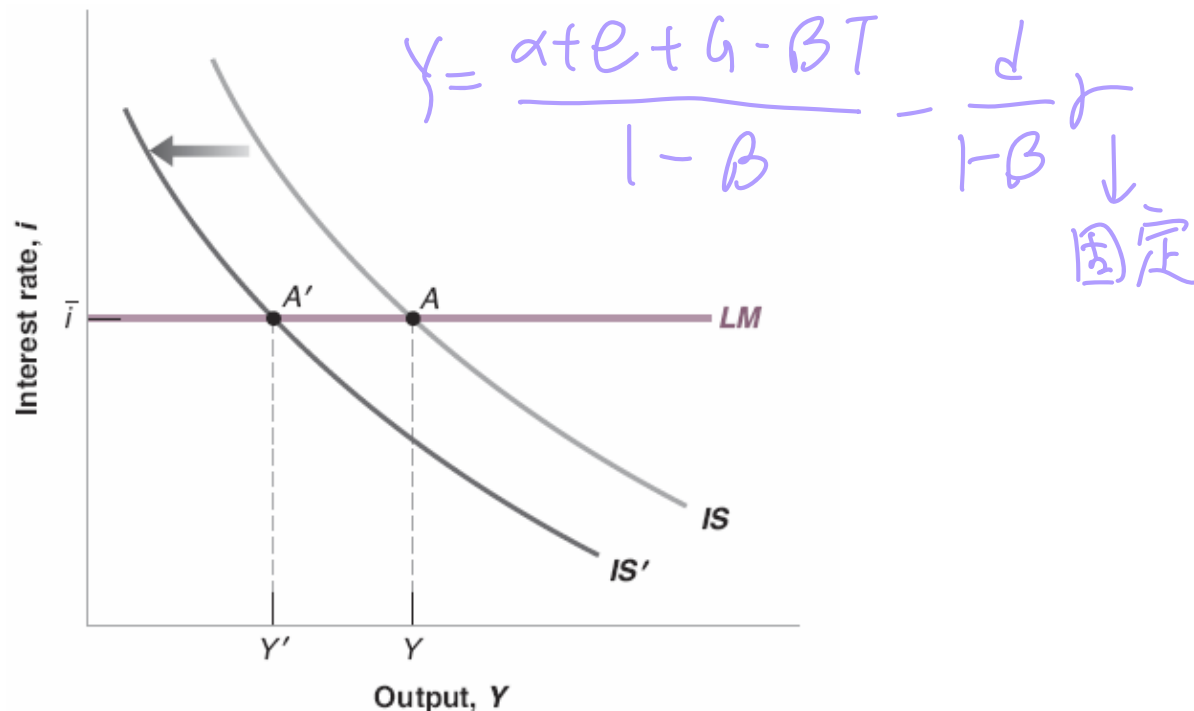
- Steps for analyzing the effects of changes in policy or exogenous variables:
 1. Does it shift the *IS* curve and/or the *LM* curve?
 2. What does this do to equilibrium output and the equilibrium interest rate?
 3. Describe the effects in words.

5.3 Putting the *IS* and the *LM* Relations Together (4 of 6)

Figure 5.6 The Effects of an Increase in Taxes

An increase in taxes shifts the *IS* curve to the left.

This leads to a decrease in the equilibrium level of output.



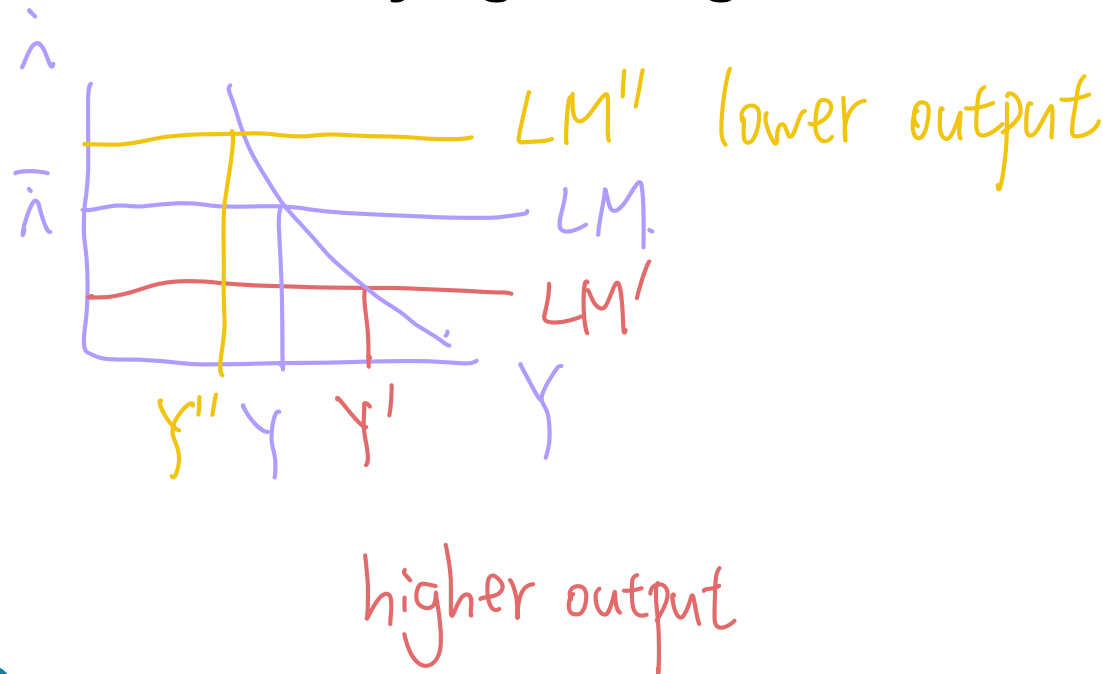
5.3 Putting the *IS* and the *LM* Relations Together (5 of 6)

- Monetary Policy:

Decrease in $i \Leftrightarrow$ increase in $M \Leftrightarrow$ **monetary expansion**

Increase in $i \Leftrightarrow$ decrease in $M \Leftrightarrow$ **monetary contraction**

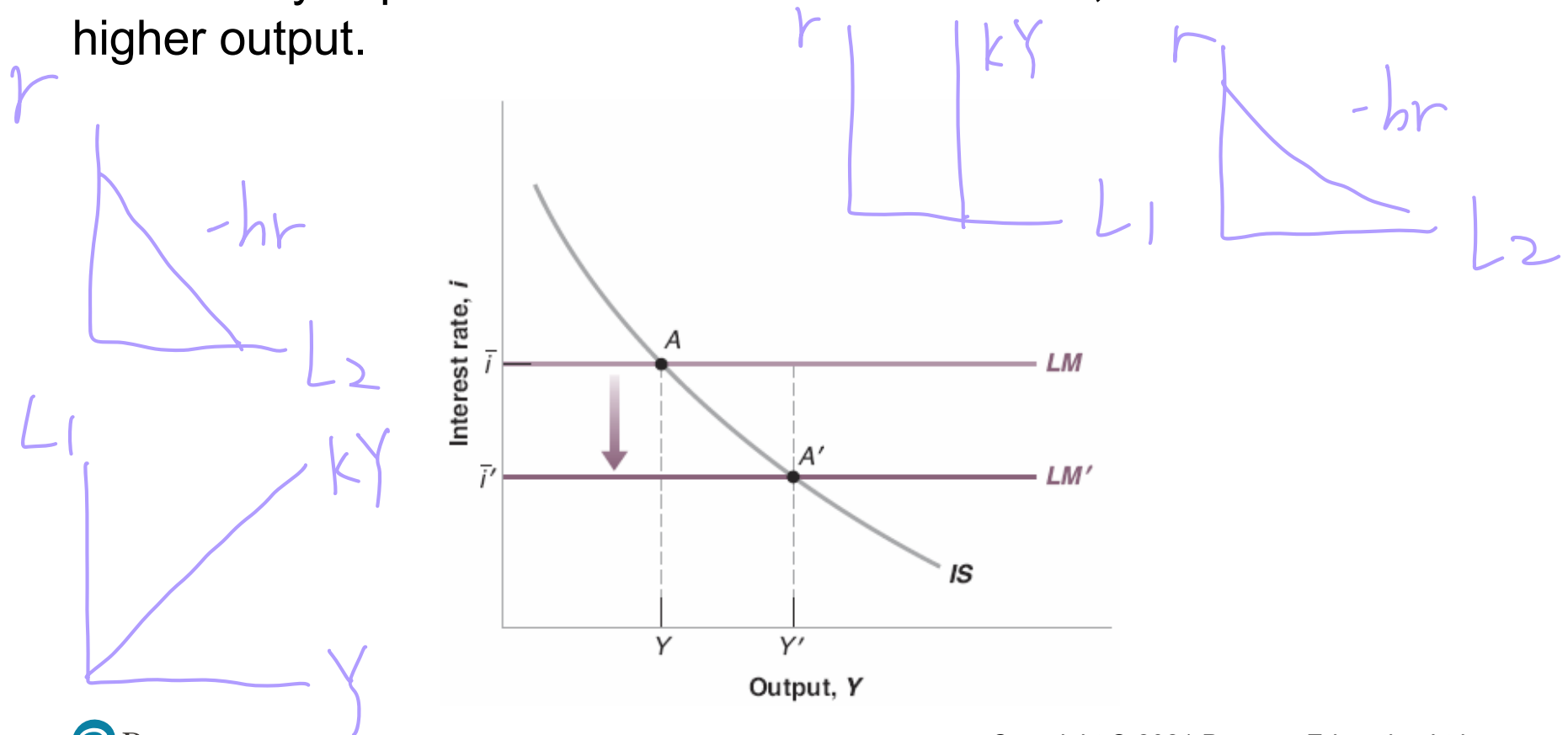
$\Leftrightarrow$ **monetary tightening**



5.3 Putting the *IS* and the *LM* Relations Together (6 of 6)

Figure 5.7 The Effects of a Decrease in the Interest Rate

A monetary expansion shifts the *LM* curve down, and leads to higher output.



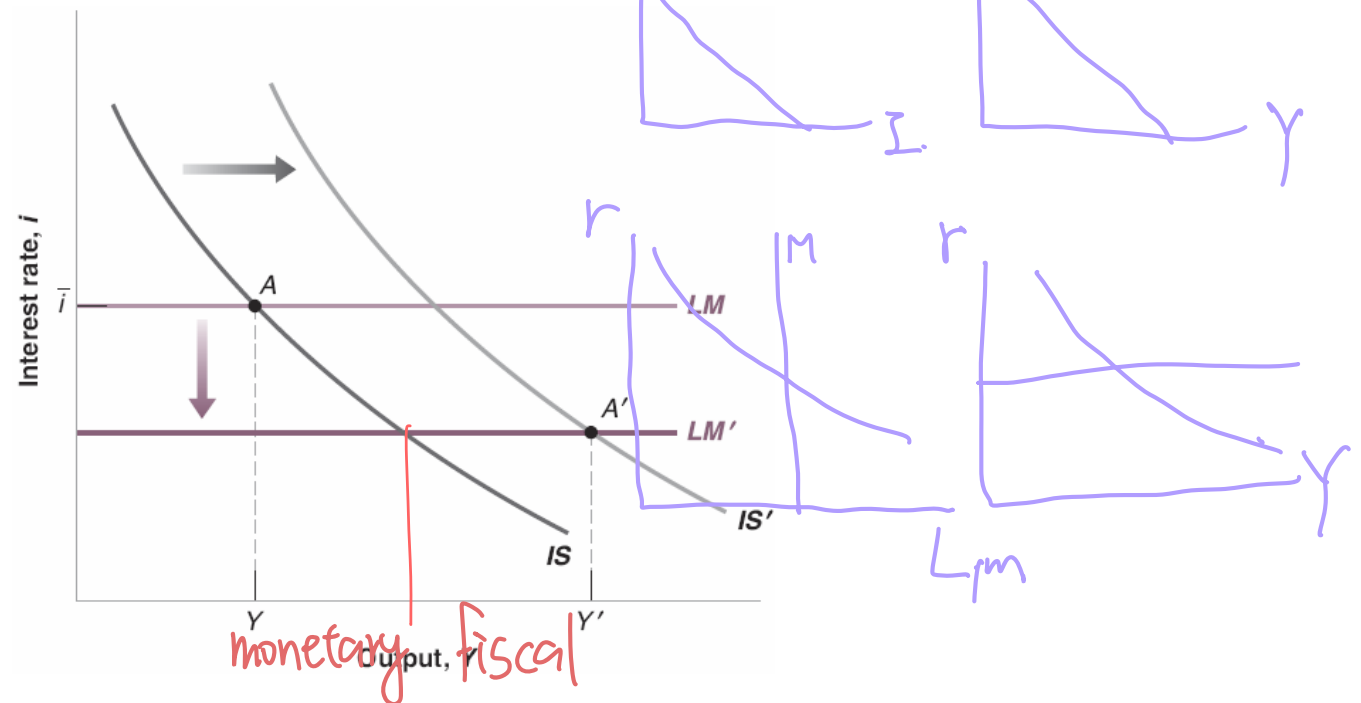
5.4 Using a Policy Mix (1 of 3)

- Monetary-fiscal policy mix is the combination of monetary and fiscal policies.
- Suppose that the economy is in a recession and output is too low.
- Both fiscal and monetary policies can be used to increase output.

5.4 Using a Policy Mix (2 of 3)

Figure 5.8 The Effects of a Combined Fiscal and Monetary Expansion

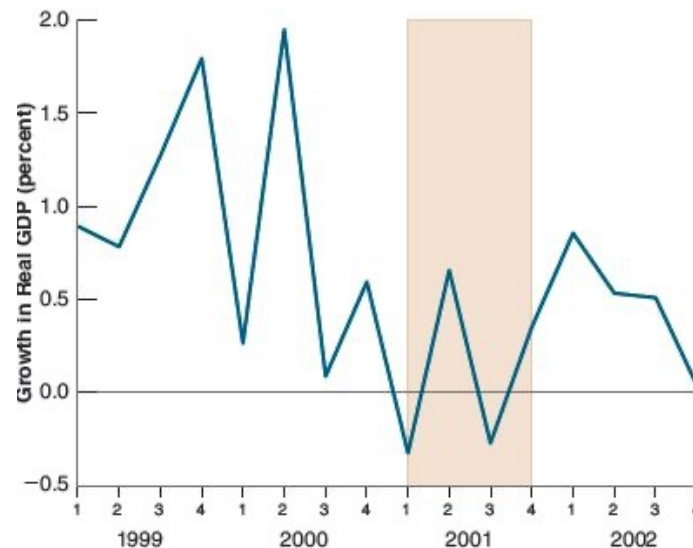
The fiscal expansion shifts the IS curve to the right. A monetary expansion shifts the LM curve down. Both lead to higher output.



Focus: The U.S. Recession of 2001 (1 of 3)

Figure 1 The U.S. Growth Rate, 1999 Q1 to 2002 Q4

The National Bureau of Economic Research concluded that the U.S. economy was in a recession between March 2001 and December 2001, triggered by sharp declines in investment demand.



Source: Calculated using Series GDPC1, Federal Reserve Economic Data (FRED)

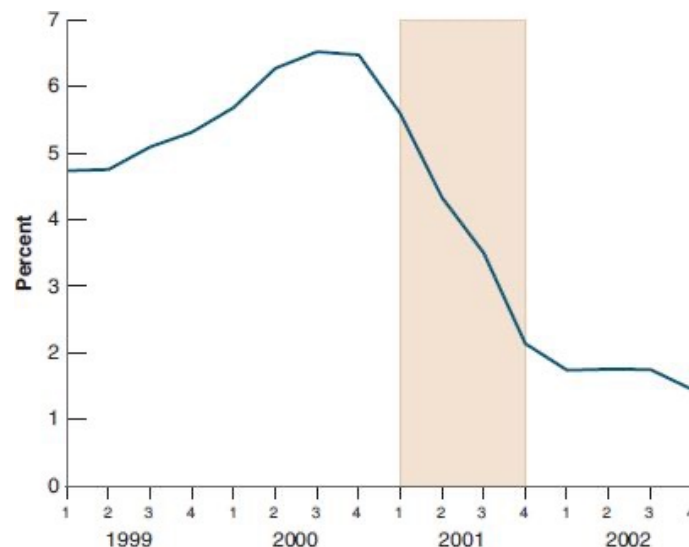
<http://research.stlouisfed.org/fred2/>

Focus: The U.S. Recession of 2001 (2 of 3)

Figure 2 The Federal Funds Rate, 1999 Q1 to 2002 Q4

The recession was met by strong macroeconomic policy response.

The Fed cut the federal funds rate from 6.5% in January to 2% at the end of 2001.

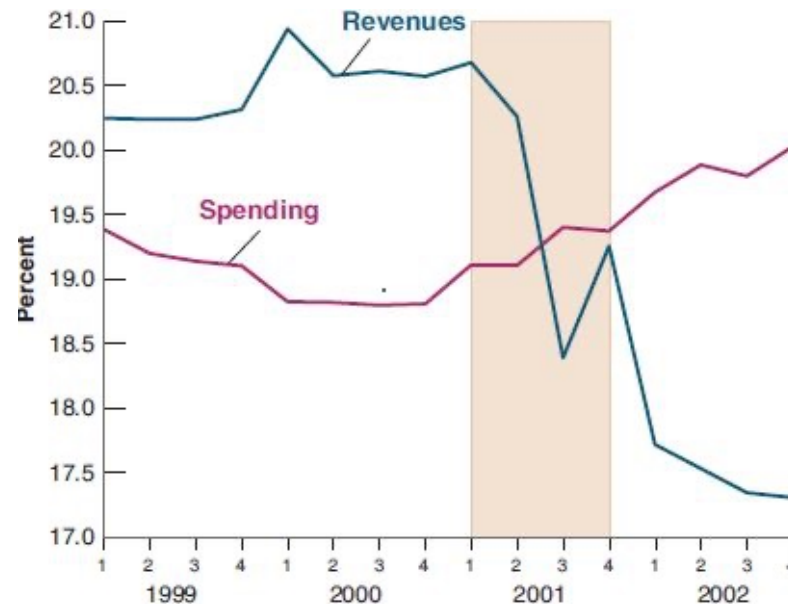


Source: Calculated using Series GDP, FGRECPY, FGEXPND, Federal Reserve Economic Data (FRED) <http://research.stlouisfed.org/fred2/>

Focus: The U.S. Recession of 2001 (3 of 3)

Figure 3 U.S. Federal Government Revenues and Spending (as Ratios to GDP), 1999 Q1 to 2002 Q4

President George Bush also cut taxes in 2001 and 2002 budgets. The events of September 11, 2001 also lead to an increase in spending on defense and homeland security.



Source: Calculated using Series GDP, FGRECPY, FGEXPND, Federal Reserve Economic Data (FRED) <http://research.stlouisfed.org/fred2/>

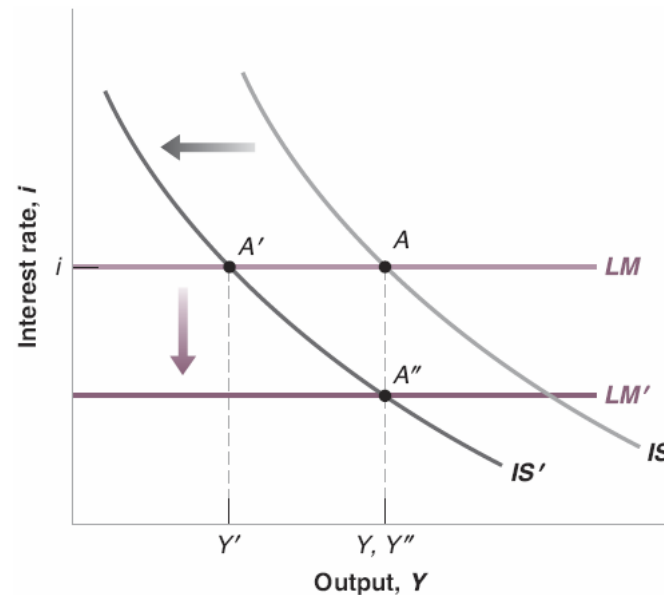
5.4 Using a Policy Mix (3 of 3)

Figure 5.9 The Effects of a Combined Fiscal Consolidation and a Monetary Expansion

The fiscal consolidation shifts the IS curve to the left.

A monetary expansion shifts the LM curve down.

This allows for the reduction in the deficit without a recession.



FOCUS: Deficit Reduction: Good or Bad for Investment?

$$I(Y, \hat{r})$$

- Equilibrium in the goods market implies (recall Chapter 3):

$$I = S + (T - G) \quad \text{private } S = Y - C - T$$

- Given private saving (S), a lower government deficit (higher $T - G$) means higher I .

$$\text{public} = T - G$$

$$Y = C + I + G$$
$$Y - T$$

- However, a fiscal contraction lowers output and so S goes down by more than $T - G$ increases, so I decreases.

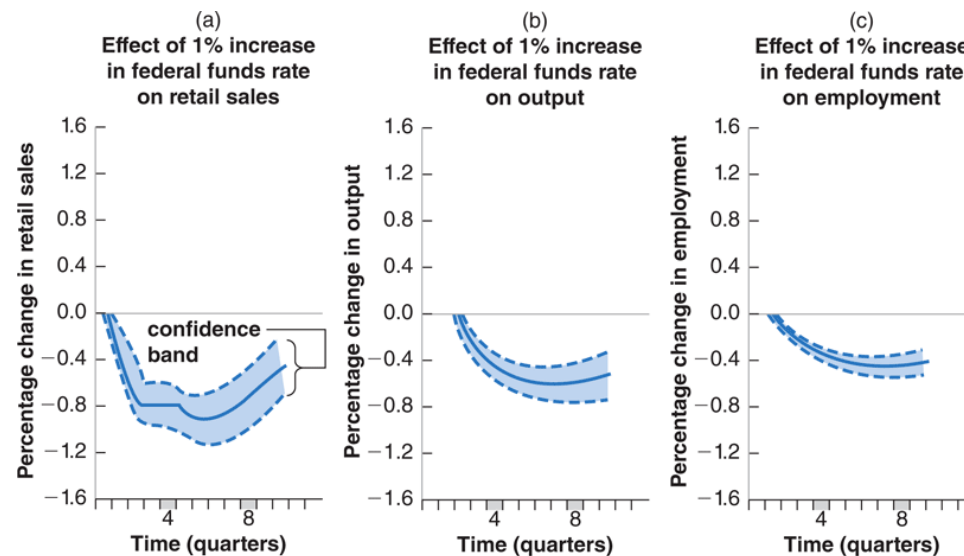
5.5 How Does the *IS-LM* Model Fit the Facts? (1 of 3)

- Because the adjustment of output takes time, we need to reintroduce dynamics:
 - Consumers are likely to take time to adjust their consumption following a change in disposable income.
 - Firms are likely to take time to adjust investment spending following a change in their sales.
 - Firms are likely to take time to adjust investment spending following a change in the interest rate.
 - Firms are likely to take time to adjust production following a change in their sales.

5.5 How Does the *IS-LM* Model Fit the Facts? (2 of 3)

Figure 5.10 The Empirical Effects of an Increase in the Federal Funds Rate

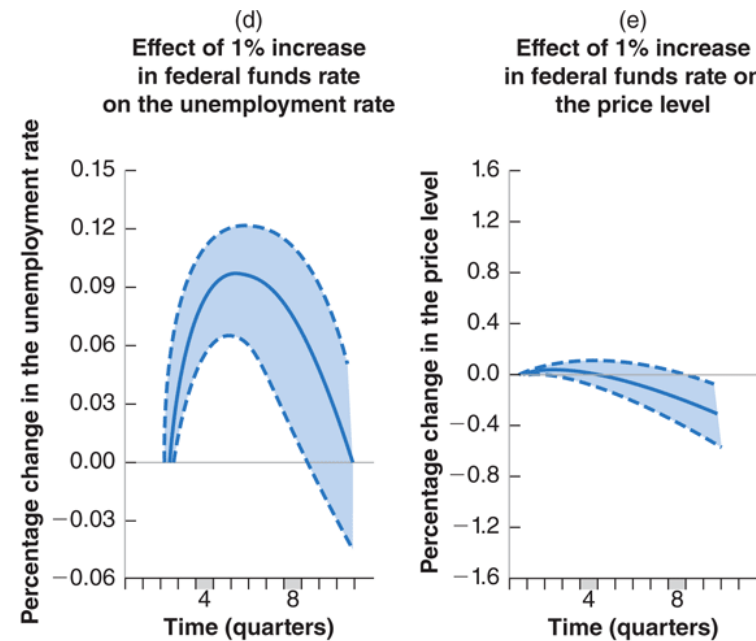
In the short run, an increase in the federal funds rate leads to a decrease in output and to an increase in unemployment, but it has little effect on the price level.



Source: Lawrence Christiano, Martin Eichenbaum, and Charles Evans, “The Effects of Monetary Policy Shocks: Evidence From the Flow of Funds,” *Review of Economics and Statistics*. 1996, 78 (February): pp. 16–34.

5.5 How Does the *IS-LM* Model Fit the Facts? (3 of 3)

Figure 5.10 The Empirical Effects of an Increase in the Federal Funds Rate



Source: Lawrence Christiano, Martin Eichenbaum, and Charles Evans, "The Effects of Monetary Policy Shocks: Evidence From the Flow of Funds," *Review of Economics and Statistics*. 1996, 78 (February): pp. 16–34.